
MA 6040 - Fuzzy Logic Connectives: Theory and Applications
Assignment 2 - Algebra of Fuzzy Sets - 29, August 2026

1. Show that the following relations are partial orders on A . Are they total order?
 - (a) $m \leq n$ iff m divides n , where $A = \mathbb{N}$.
 - (b) $m \leq n$ iff m divides n , $A =$ the set of all positive-integer divisors of 36.
 - (c) If $(A_1, \leq_1)$ and $(A_2, \leq_2)$ are two posets, then the dictionary order $\leq$ on $A = A_1 \times A_2$ is given by $(x_1, x_2) \leq (y_1, y_2)$ iff $x_1 <_1 y_1$ or $(x_1 = y_1 \text{ and } x_2 \leq_2 y_2)$. Show that $(A, \leq)$ is a poset.
 - (d) Show that the divisibility relation is not a partial order on the set of integers $\mathbb{Z}$. Which property is lacking?
2. Prove or Disprove: If $(A, \leq)$ and $(B, \leq)$ are partial ordered sets then $(A \cup B, \leq)$, $(A \cap B, \leq)$ and $(A \times B, \leq)$ are also partially ordered. If they are not, what assumptions on A, B will give us the desired result?
3. Let $(L, \vee, \wedge)$ be an algebraic lattice. If we define:

$$x \leq y \iff x \vee y = y \text{ or } x \wedge y = x$$

Then show that $(L, \leq)$ is a Lattice. Highlight for yourself how the different properties of $\wedge, \vee$ play a role in your proof.

4. Let $(L, \leq)$ be a Lattice and $x, y \in L$. Then show that the following are equivalent;
 - (a) $x \leq y$,
 - (b) $\sup(x, y) = y$,
 - (c) $\inf(x, y) = x$.
5. Let $(A, \leq)$ be a totally order set.
 - (a) Prove that $(A, \leq)$ is a lattice.
 - (b) Prove that if A has more then two elements, then it is not a complimented lattice.
 - (c) Does every lattice have maximal and minimal elements?
6. Let $(P, \vee, \wedge, ', 1, 0)$ be a Boolean algebra. Prove the following identities:
 - (a) $x'' = x$,
 - (b) $x \vee (y \vee y') = x$,
 - (c) $x \vee (y \vee z)' = ((y' \vee x)' \vee (z' \vee x)')'$
 - (d) $(x' \vee y') \vee (x' \vee y) = x$
7. Let $\mathbb{Z}$ be the set of all integers. Let m be an arbitrary integer. Define

$$\mathcal{A}_m = \{P \subseteq X \mid P + m = P\}.$$

Show that $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ are Boolean algebras.

8. Let $(S, \vee, \wedge, *, 0, 1)$ be a Stone algebra, i.e., S is a pseudo-complemented distributive (bounded) lattice. Show that $(S^*, \vee, \wedge, *, 0, 1)$ is a Boolean algebra, where

$$S^* = \{s^* \in S \mid s \in S\}.$$

S^* is called the center of S .

9. Consider the bounded distributive lattice of fuzzy sets $(\mathcal{F}(X), \vee, \wedge, 0, 1)$.

- If $A \in \mathcal{F}(X)$, how will you define the pseudo-complement A^* of A ?
- Find the center of $\mathcal{F}(X)$ w.r.to the $*$ defined by you.
- Show that $(\mathcal{F}(X), \vee, \wedge, *, 0, 1)$ is a Stone algebra whose center consists of the set of classical (crisp) sets of x .

10. Consider the following pairs of operations (T, S) :

- (a) $T_{\mathbf{M}}(x, y) = \min(x, y), \quad S_{\mathbf{M}}(x, y) = \max(x, y),$
- (b) $T_{\mathbf{P}}(x, y) = xy, \quad S_{\mathbf{P}}(x, y) = x + y - xy,$
- (c) $T_{\mathbf{LK}}(x, y) = \max(0, x + y - 1), \quad S_{\mathbf{LK}}(x, y) = \min(1, x + y),$
- (d) $T_{\mathbf{D}}(x, y) = \begin{cases} \min(x, y), & \text{if } \max(x, y) = 1, \\ 0, & \text{otherwise.} \end{cases} \quad S_{\mathbf{D}}(x, y) = \begin{cases} \max(x, y), & \text{if } \min(x, y) = 0, \\ 1, & \text{otherwise.} \end{cases}$

Seeing them as binary operations on $[0, 1]$ discuss the following:

- Are all of these operations associative?
 - Are the pairs (T, S) distributive over each other?
 - For which pairs (T, S) can you find an appropriate complement / pseudo-complement operation?
 - Which of these operations satisfy(ies) the Stone's identity?
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